



CS 540 Module One Assignment Guidelines and Rubric

Overview

Before using data to make decisions, develop visualizations, or support recommendations, it's essential to ensure the data set is well-organized, consistent, and suitable for analysis. This skill is vital across many professional fields. Whether you're building dashboards, conducting research, or supporting leadership with data-driven recommendations, your ability to evaluate and refine data will strengthen the quality and credibility of your work.

Directions

In this assignment, you'll practice examining a data set to evaluate its quality by focusing on how clean, complete, and logically structured it is. You'll also assess how useful the data set is for answering real-world questions and determine what additional data might improve the overall analysis. Before you begin, choose a data set from the Module One Assignment data folder in Codio. When you are ready to begin your work, access Codio by clicking the title of the next task.

A sample Python Jupyter Notebook with pre-built exploratory data analysis content has been provided as a tool in your evaluation of the data quality. The data can be downloaded for evaluation via other tools of preference.

Specifically, you must address the following rubric criteria:

1. Describe your **selected data set**. Include the following:
 - A. The source of the data set
 - B. An explanation of whether the source is reputable
 - C. The data types used in each column
2. **Identify** any data quality issues that could affect analysis. Include the following:
 - A. Missing or null values, inconsistent formats, mislabeled columns, or unusual entries
3. Evaluate the overall **quality and cleanliness** of the data. Include the following:
 - A. Support your conclusion with specific examples.
4. Explain how the **data quality issues** might impact your ability to analyze the data effectively.
5. Determine what kinds of **questions or insights** this data set could help answer.

What to Submit

Submit your assignment as a 2- to 3-page Microsoft Word document with double spacing, 12-point Times New Roman font, and one-inch margins.

AI Usage

If you use gen AI tools to support your work on this assignment, be sure to follow [these AI usage guidelines](#). You must acknowledge your use of these tools in your work. Guidelines on how to cite AI tools can be found in [this Shapiro Library guide](#).

Module One Assignment Rubric

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Selected Data Set	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Describes the selected data set, including the source of the data set, an explanation of whether the source is reputable, and the data types used in each column	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Identify	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Identifies any data quality issues that could affect analysis, including missing or null values, inconsistent formats, mislabeled columns, or unusual entries	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Quality and Cleanliness	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Evaluates the overall quality and cleanliness of the data, including supporting the conclusion with specific examples	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Data Quality Issues	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Explains how the data quality issues might impact the ability to analyze the data effectively	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19
Questions or Insights	Exceeds expectations in an exceptionally clear, organized, insightful, or sophisticated manner	Determines what kinds of questions or insights this data set could help answer	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	19

Criteria	Exceeds Expectations (100%)	Meets Expectations (90%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Clear Communication	Exceeds expectations with an intentional use of language that promotes a thorough understanding	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	5
Total:					100%